

1.a) Define Computing. Explain all the major innovations in the field of hardware, software and networking.

sol:- Computing is the process of using computers or other devices to perform tasks such as processing data, solving problems and executing instructions.

The history of computing has witnessed several major breakthroughs across hardware, software and networking. Some of the most significant innovations in each of these areas are as follows:-

* Hardware Innovations;

→ The major innovations in the field of hardware are as follows:-

(i) Vacuum Tubes (1940s): Early electronic computers used vacuum tubes for processing.

(ii) Transistors (1950s): Transistors replaced vacuum tubes because they were smaller, faster and more reliable in comparison.

(iii) Integrated Circuits (1960s): Integrated circuits combined many transistors on a single chip which became more powerful as compared to previous transistors.

(iv) Microprocessors (1970s): Microprocessors combined entire CPU on a single chip (Intel 4004) which became very popular as compared to previous.

(v) Personal Computers (1980s): Personal Computers made computing accessible to every individuals for their offices ~~and~~, homes and so on.

(vi) Modern Hardware (2000s - now): Today modern hardware like multi-core processors, GPUs, mobile devices, IoT hardware etc become very popular.

* Software Innovations;

→ The major innovations in the field of software are as follows:-

- (i) Machine Languages: Early computers were programmed using binary code i.e. machine language.
- (ii) Assembly Languages: Assembly languages provided a more human-readable representation of machine instructions, making programming easier.
- (iii) High-Level Programming Languages: The development of HLL like FORTRAN, COBOL, LISP, and C simplified the programming tasks.
- (iv) Operating Systems: OSs like UNIX, Windows, made it easier to use computers.
- (v) Graphical User Interface (GUIs): Macintosh and Microsoft Windows introduced GUIs which made computers more user-friendly and intuitive.
- (vi) Modern Software: Modern softwares like open-source software, mobile apps, cloud computing, AI applications which helps innovation grow faster.

* ~~So~~ Networking Innovations;

→ The major innovations in the field of networking are as follows:-

- (i) ARPANET (U.S.): ARPANET was created by the U.S. Department of Defense's Advanced Research Projects Agency (ARPA). It was the first network to use the TCP/IP protocol suite.
- (ii) Ethernet: Ethernet became the dominant local area network (LAN) technology, providing fast and reliable data transmission.

(iii) IP (Internet Protocol): The IP is the addressing system of the internet. Every device on the internet needs an address so that data knows where to go. IP has two versions i.e. IPv4 (32-bit address) and IPv6 (128-bit address).

(iv) DNS (Domain Name System): DNS allowed the mapping of human-readable name to numerical IP addresses, making it easier for users to access websites.

(v) WWW (World Wide Web): WWW brought a user-friendly and widely accessible interface to the internet, making it easy to navigate and use.

1. b) What do you understand by "Social Context of Computing"? Explain the impacts of technology on society and vice-versa.

Ans:

- Social Context of Computing means understanding how computing technologies affect society and how society shapes the development and use of technology.
- It is concerned with the impact of computers on individuals, communities and organizations.
- It includes ethical issues like privacy, security and digital rights.

The impacts of technology on society and vice-versa are as follows:-

① Impacts of Technology on Society:

→ Technology has changed the way we live, work, learn and communicate. Some of the impacts are as follows:-

(i) Better Communication:

→ Technology has made communication faster, easier and global. People can stay connected with family, friends and colleagues no matter where they are.

(ii) Improved Healthcare:

→ Modern medical equipment, telemedicine and health apps help in faster diagnosis, better treatment and even remote care.

(iii) Education Access:

→ Technology has brought learning to everyone's fingertips. Online classes, digital libraries and e-learning platforms let people learn anytime, anywhere.

(iv) Job Creation and Automation;

→ Technology creates new types of jobs (like app developers, data analysts, cybersecurity experts) but also replaces some jobs through automation (like robots in factories).

(v) Economic Impact:

→ Technology has driven economic growth through the creation of new industries and business models.

(vi) Environmental Impact:

→ While technology has helped in finding solutions (like clean energy), it also contributes to pollution and e-waste if not managed properly.

② Impacts of Society on Technology:

→ Society also shapes technology by following impacts:
(i) ~~Society's needs and demands shape the development of technology.~~

(i) User Needs and Demands:

→ Society's needs and demands shape the development of technology.

(ii) Ethical and Legal Considerations:

→ Societal values and ethical concerns influence how technology is designed, used and regulated.

(iii) Cultural Influence:

→ Cultural factors influence the adoption and use of technology.

(iv) Digital Divide:

→ ~~Societal disparities in access to technology and digital literacy can affect its adoption and success.~~

(iv) Digital Divide:

→ Societal disparities in access to technology and digital literacy can impact how technology is implemented and whom it serves.

(v) Economic Factors:

→ Economic considerations play a role in determining which technologies are viable and how they are commercialized.

2.a) What is Ethical Relativism? What do you mean by Utilitarianism and Deontological theories? Explain.

Eq: Ethical Relativism is a philosophical theory that posits ethical principles and moral judgements ~~to~~ relative are relative to the cultural, historical or individual context in which they occur.

In other words, what is right or wrong depends on the culture, society or individual beliefs not on universal moral rules.

According to ethical relativism, different societies or people can have different views on what is moral, and no one view is absolutely correct for everyone.

For example:

In one culture, eating a certain type of food may be acceptable, while in another, it may be seen as wrong both are right in their own context.

(*) Utilitarianism: Theories:

- Utilitarianism is an ethical theory that says an action is right if it brings the greatest happiness or benefit to the greatest number of people.
- It focuses on the consequences of actions. If an action produces more good than harm overall, it is considered ethical.

(*) Deontological Theories:

- Deontological theories are ethical theories that say actions are right or wrong based on rules, duties or principles, not just on the results.
- It focuses on doing your duty and following moral rules matters than the outcome.

Let's consider ~~the~~ an ~~exp~~ example of the practice of capital punishment.

Utilitarianism would evaluate whether the death penalty promotes overall happiness or reduces suffering in a society. If it is deemed to deter crime and protect citizens, a utilitarian might support its use.

On the other hand, a deontologist might focus on the moral principle that killing another human being is inherently wrong, regardless of the potential benefits. Therefore, a deontologist would likely oppose the death penalty based on the principle of preserving human life.

2.b) With help of examples, discuss how each is different among the terms intellectual property, patent trademark, copy right.

Sol:

(i) Intellectual Property:

→ An intellectual property is a general term that refers to creations of the mind: ideas, inventions, designs, software, writings etc.

→ It gives the creator legal rights to protect and control how their creation is used.

→ Example:

A company's logo, a new machine design, a song or a book are all types of intellectual property.

(ii) Patent:

→ ~~A patent is an invention~~

→ A patent is a legal right given for an invention i.e. something new, useful and not obvious.

→ It protects how the invention works and stops others from making, using or selling it without permission.

→ For example:

The design of a new engine, a new type of mobile phone, or a new medicine formula can be patented.

(iii) Trademark:

→ A trademark protects symbols, names, logos, or slogans that identify and distinguish a brand's goods or services.

→ It stops others from using a similar mark that could confuse customers.

→ For example:

The word 'Coca-Cola', The Nike logo, The Adidas logo, etc.

(iv) Copyright :

→ A copyright protects original creative works like books, music, films, art and software.

→ It gives the creator the right to copy, distribute, perform or display the work.

→ Example;

A novel, a song, a movie or a computer program is protected by copyright.

* Summary of differences between them:

Term	Meaning	Example:
Intellectual Property	General term for creations of the mind.	A brand's logo, a book, a software, a machine design.
Patent	New inventions and how they work.	A new smartphone technology or a vaccine formula.
Trademark	Brand names, logos, slogans.	The Apple logo, Nike wordmark, adidas logo etc.
Copyright	Original creative and artistic works.	A movie, a song, a painting, a software code.

3.a) What is code of ethics? Mention any five code of ethics proposed by NEC?

Sol :-

A code of ethics is a set of principles designed to guide professionals in their work, ensuring integrity, responsibility and fairness.

It serves as a framework for ethical decision-making and behavior, promoting integrity, responsibility and accountability in the profession.

The Nepal Engineering Council (NEC) is a regulatory body that governs the engineering profession in Nepal. It has established a code of Ethics to guide engineers in their practice and ensure ethical conduct.

Some code of ethics proposed by NEC are as follows:-

(i) Discipline and Honesty:

→ The engineering profession must be conducted in a disciplined manner with honesty, not contravening professional dignity and well-being.

(ii) Politeness and Confidentiality:

→ Engineering services for customers should be dealt with ~~with~~ in a polite manner and professional information should remain confidential except with written or verbal consent of the customers concerned.

(iii) Non-discrimination:

→ No discrimination should be made against customers on the grounds of region religion, race, sex, caste, or any other things while applying professional knowledge and skills.

(iv) Professional Work:

→ Individuals should only do professional work in their field or provide recommendations or suggestions only within the area of their subject of study or obtained knowledge or skills.

(v) Personal Responsibility:

→ All individuals will be personally responsible for all works performed in connection with his/her engineering profession.

(vi) State name, designation and registration No.:

→ While signing the documents related to the engineering profession, the details should include, the name, designation and NEC registration number and should be stated in a clear and comprehensible manner.

(vii) No publicity or advertisement must be made which may cause unnecessary effect:

→ In connection with the professional activities to be carried out, no publicity or advertisement shall be made so as to cause unnecessary effect upon the customers.

3.6) Differentiate between descriptive and normative claims with example.

sol) Descriptive and normative claims are two types of statements used in philosophy, ethics and social sciences. Their differences are as follows:-

S.N.	Descriptive claims	Normative claims
(i)	Descriptive claims state facts or describe how things are or were.	Normative claims express judgements about how things ought to be or should be.
(ii)	Descriptive claims are so objective in nature.	Normative claims are subjective in nature.
(iii)	Descriptive claims focus on observation or reality.	Normative claims focus on values, ethics or moral beliefs.
(iv)	Descriptive claims have purpose to explain or report what actually happens or exists.	Normative claims have purpose to prescribe, recommend or evaluate actions or situations.
(v)	Descriptive claims are based on facts, evidence or reality.	Normative claims are ^{based on} moral standards, values or principles.
(vi)	Descriptive claims are testable because as it can be verified or proven true/false.	Normative claims are not testable as it depends on ethical or moral views.
(vii)	Example: Many people use social media daily.	Example: People should limit their use of social media to protect mental health.

4.a) Define safety. Why safety in critical system is necessary?

Def: Safety is the condition of being protected from harm, danger or risk.

→ In critical systems, safety means designing and operating a system in a way that prevents accidents or failures that could cause injury, loss of life, damage to property or harm to the environment.

→ Critical systems are systems where where failure can lead to serious consequences, such as loss of life, major financial loss, or environmental damage.

→ Examples include: aircraft control systems, medical devices, nuclear power plants and railway signalling systems.

In critical systems, safety is very necessary because of the following reasons:

(i) To prevent loss of life or injury:

→ Failure of critical systems can result in accidents that harm or kill people (e.g. failure of an aircraft's autopilot system).

(ii) To protect property and infrastructure;

→ Failure of critical systems could destroy expensive equipment or critical infrastructure (e.g. collapse of a bridge due to faulty design software).

(iii) To avoid environmental disasters;

→ Critical systems like nuclear power plants need high safety to prevent harm to the environment (e.g. radiation leaks).

(iv) To ensure public trust and confidence;

→ People must feel safe using services that rely on critical systems (e.g. trusting the safety of elevators, planes or trains).

(v) To meet legal and ethical responsibilities;

→ There are laws and regulations requiring safety in critical systems and engineers have a moral duty to design safe systems.

In short, safety in critical systems is necessary to prevent harm, protect people and the environment and ensure reliable and responsible use of technology.

4.b) What do you mean by values in design? Explain hardware and software design issues.

Def: Values in design means integrating ethical, social and human values (e.g. privacy, accessibility, safety) into technology design.

→ It ensures technology is not just efficient or powerful but also safe, fair, accessible, environmentally friendly and so on.

⇒ In short, values in design help build technology that is good not just for business only, but for people and society.

(*) Hardware Design Issues:

→ While designing hardware, engineers must think beyond technical performance. Some of the significant hardware design issues are as follows:

(i) Performance:

→ Designers need to optimize the hardware's performance to meet the desired processing speed, bandwidth and power efficiency requirements.

(ii) Power Consumption:

→ Power efficiency is crucial for many devices, particularly in mobile devices and battery-operated systems.

(iii) Heat Dissipation:

→ As hardware components become more powerful, managing heat dissipation becomes increasingly important to prevent overheating.

(iv) Compatibility:

→ Ensuring compatibility between hardware and components and various software is essential for seamless user experiences.

(iv) Scalability:
→ Designers must consider how well the hardware can scale to handle increasing workloads or accommodate future upgrades.

(v) Reliability:
→ Hardware Components should be reliable and have a long operational lifespan.

(vi) Cost:
→ Designers need to balance performance and features with the cost of manufacturing to produce products.

(*) Software Design Issues:

→ Software design also carries following responsibilities:

(i) Privacy:

→ Software should protect user data from misuse or leaks.

(ii) Fairness:

→ Software algorithms must avoid bias, providing fairness.

(iii) Reliability:

→ Software must be reliable, working as expected especially in critical areas.

(iv) Security:

→ Software should resist hacking and protect sensitive information.

(v) User-friendly:

→ Software should be designed so all users can use it easily including those with disabilities.

(vi) Transparency and Accountability:

→ It should be clear how software works, especially in critical systems.

6. a) Differentiate between E-Governance and E-Government

So

S.N.	E- Government	E- Governance
(i)	E- Government means using technology to provide government services to people online	E- Governance means using technology to improve government working, decision-making and public participation.
(ii)	E- Government has narrower scope which focuses mainly on service delivery.	E- Governance has broader scope which covers decision-making, policies, participation and service delivery.
(iii)	E- Government focuses on providing online government services (e.g. tax filling, license renewal).	E- Governance focuses on improving government processes, engaging citizens, making governance more transparent.
(iv)	E- Government includes: G2C, G2B, and G2G services.	E- Governance includes: G2C and C2G and other interactions.
(v)	E- Government provides efficiency in delivering services.	E- Governance provides better governance, accountability, transparency and citizen participation.
(vi)	E- Government has limited citizen participation.	E- Governance has higher citizen participation.
(vii)	Example: Online passport application system, e-tax filling portal.	Example: Giving suggestions on a public portal, filing complaints online, digital voting.